

Enthusiast Course

PHASE : SPEED & ACCURACY TEST

TARGET : PRE-MEDICAL 2022

Test Type : PRACTICE TEST

Test Pattern : NEET (UG)

TEST DATE :

ANSWER KEY

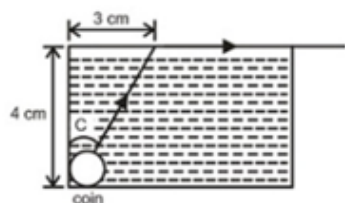
Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A.	4	2	1	4	2	1	1	3	1	3	4	1	2	1	1	2	3	2	1	1	1	1	4	3	2	1	4	2	3	2
Q.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45															
A.	2	4	3	3	1	2	1	1	4	2	2	4	3	1	1															



Click Here To Join our Telegram Channel

HINT - SHEET

1. Ans (4)



$$\text{From figure, } \sin C = \frac{3}{\sqrt{4^2 + 3^2}} = \frac{3}{5}$$

Where C is the critical angle.

$$\text{Also, } \sin C = \frac{1}{\mu_1}$$

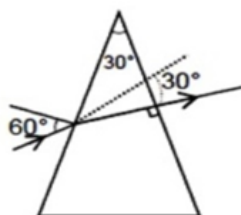
$$\mu_1 = \frac{\text{velocity of light in air 'c'}}{\text{velocity of light in liquid 'v'}}$$

$$\sin C = \frac{v}{c} = \frac{v}{3 \times 10^8}$$

or

$$v = 3 \times 10^8 \times \frac{3}{5} = 1.8 \times 10^8 \text{ m/s}$$

2. Ans (2)



$$\delta = i + e - A$$

$$30 = 60 + e - 30$$

$$e = 0$$

So angle with face = 90°

3. Ans (1)

For a real inverted image formed by concave

mirror v , u , and f all are negative and

$\frac{u}{f}$ and $\frac{v}{f}$ are positive so graph should be in

1st quadrant and

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f} \Rightarrow \frac{f}{v} + \frac{f}{u} = 1$$

This graph can be also realized by Newton's

formula $x_1 x_2 = f^2$ where x_1 and x_2 are

distance from focus so graph will be

rectangular hyperbola.

4. Ans (4)

By using lens formula

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{v} = \frac{1}{0.2} + \frac{1}{-0.3}$$

$$v = 60 \text{ cm}$$

5. Ans (2)

From displacement method

$D = 90 \text{ cm}$ and $d = 20 \text{ cm}$

$$F = \frac{D^2 - d^2}{4D} = 21.4 \text{ cm}$$

6. **Ans (1)**

$$\frac{\omega_1}{\omega_2} = \frac{1}{2}$$

Now,

$$\frac{f_1}{f_2} = -\frac{\omega_1}{\omega_2} = -\frac{1}{2}$$

$$\text{Or } f_2 = -2 f_1$$

Now,

$$\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2} \quad \text{Tele: @chalnaayaaaar_bot}$$

$$\frac{1}{50} = \frac{1}{f_1} + \frac{1}{-2f_1}$$

$$\frac{1}{50} = \frac{-2+1}{-2f_1} = \frac{1}{2f_1}$$

$$2f_1 = 50 \text{ or } f_1 = 25 \text{ cm}$$

$$\text{Again } f_2 = -2 \times 25 \text{ cm}$$

$$f_2 = -50 \text{ cm}$$

7. **Ans (1)**

We have to consider sign convention as only magnitude of focal length is given.

If f_1 is focal length of convex (converging) lens and f_2 is focal length of concave (Diverging) lens. Then their equivalent focal length F would be

$$\left[\because \frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2} \right]$$

$$\frac{1}{F} = \frac{1}{f_1} + \frac{1}{-f_2} = \frac{f_2 - f_1}{f_1 f_2}$$

$$\therefore F = \frac{f_1 f_2}{f_2 - f_1}$$

8. **Ans (3)**

Given,

$$\mu_g = \mu_l$$

The focal length of convex lens in liquid f is given by

$$\frac{1}{f} = \left(\frac{\mu_g}{\mu_l} - 1 \right) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

$$\frac{1}{f} = (1 - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

$$\frac{1}{f} = 0$$

$$\text{Or } f = \infty$$

Its focal length will become infinite.

9. **Ans (1)**

$$\text{Differentiating the lens formula } \frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

with respect to time, we get

$$-\frac{1}{v^2} \cdot \frac{dv}{dt} + \frac{1}{u^2} \cdot \frac{du}{dt} = 0;$$

$$\left(\text{as } f = \text{constant} \right)$$

$$\therefore \frac{dv}{dt} = \left(\frac{v^2}{u^2} \right) \cdot \frac{du}{dt} \quad \text{--- (1)}$$

Further, substituting proper values in lens formula, we

$$\frac{1}{v} + \frac{1}{0.4} = \frac{1}{0.3} \quad (u = -0.4 \text{ m}, f = 0.3 \text{ m})$$

$$\text{or, } v = 1.2 \text{ m}$$

Putting the values in eq (i) magnitude of rate of change of position of image = 0.09 m/s

$$\text{Lateral magnification, } m = \frac{v}{u}$$

$$\begin{aligned} \therefore \frac{dm}{dt} &= \frac{u \cdot \frac{dv}{dt} - v \cdot \frac{du}{dt}}{u^2} \\ &= \frac{(-0.4)(0.09) - (1.2)(0.01)}{(0.4)^2} \\ &= -0.3/\text{s} \end{aligned}$$

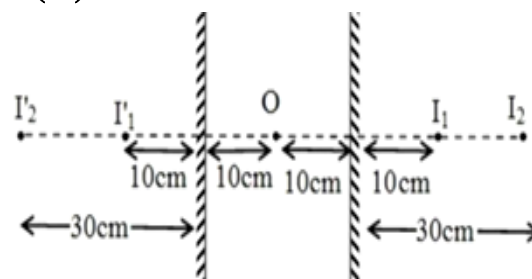
11. **Ans (4)**

Apparent height of the flame above water surface.

$$h' = \mu h = \frac{4}{3} \times 2 = \frac{8}{3} \text{ m}$$

$$\text{Therefore, apparent height of the flame from the eye of the fish} = d + h' = 4 + \frac{8}{3} = \frac{20}{3} \text{ m}$$

12. **Ans (1)**



From the diagram it is clear that distance between I_2 & $I_2' = (30 + 20 + 30) \text{ cm} = 80 \text{ cm}$

13. **Ans (2)**

Angular magnification of telescope is 20. So tree appears to the observer is 20 times nearer.

14. **Ans (1)**

Our eye lens has a power to adjust its focal length to see the nearer and farther objects, this process of adjusting focal length is called accommodation. However, if the object is brought too close or brought too far from the eye, the focal length cannot be adjusted to form the image on the retina. Thus, there is minimum or maximum distance for the clear vision of an object. For a normal eye, near point or least distance of distinct vision $D = 25$ cm and far point $= \infty$.

15. **Ans (1)**

$$n = \frac{360^\circ}{\theta^\circ} - 1 = \frac{360^\circ}{60^\circ} - 1 = 5$$

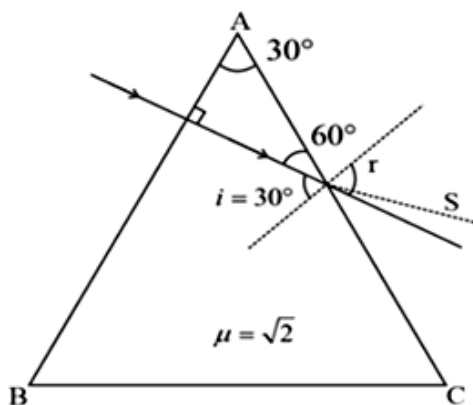
16. **Ans (2)**

Along normal to the mirror

$$\vec{v}_0 + \vec{v}_i = 2\vec{v}_m \Rightarrow \vec{v}_{ix} = 0$$

Components of velocity parallel to the mirror remain same.

17. **Ans (3)**



Using Snell's law for the refraction at AC, We get

$$\mu \sin i = (1) \sin r$$

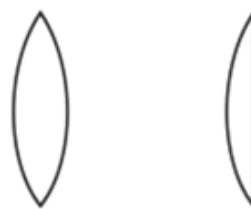
$$\sqrt{2} \sin 30^\circ = \sin r$$

$$\sin r = \frac{1}{\sqrt{2}}$$

$$r = 45^\circ$$

$$\therefore \text{Angle of deviation} = 45^\circ - 30^\circ = 15^\circ$$

18. **Ans (2)**



$$\frac{1}{f} = (\mu - 1) \left(\frac{2}{R} \right) \frac{1}{f'} = (\mu - 1) \left(\frac{1}{R} \right)$$

$$f' = 2f$$

So, cutting a lens in transverse direction doubles their focal length but if a lens is cut across its principal axis, then there is no change in focal length. So, all the 4 parts will have same focal length i.e., $2f$

Using the formula of equivalent focal length, we can find net focal length

$$\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2} + \frac{1}{f_3} + \frac{1}{f_4}$$

19. **Ans (1)**

Refractive index

$$\mu = \frac{\text{real depth}}{\text{apparent depth}}$$

Refractive index is maximum for violet colour and minimum for red colour. Hence letter which appears minimum raised is red.

20. **Ans (1)**

The given lens is a convex lens. Let the magnification be m , then

$$m = \frac{f}{f + u}$$

$$m_x = -m = \frac{f}{f - x} \dots \dots \dots (i)$$

And for virtual image

$$m_y = +m = \frac{f}{f - y} \dots \dots \dots (ii)$$

From equation (i) and (ii),

We get

$$f = \frac{x + y}{2}$$

21. **Ans (1)**

Magnifying power of a telescope is $\frac{f_0}{f_e} = 10$

Length of telescope is $f_0 = f_e = 1.1$ m

So, $f_0 = 1$ m & $f_e = 0.1$ m

When the final image is at D; then magnification is

$$m = f_0 \left(\frac{1}{D} + \frac{1}{f_e} \right) = 100 \left(\frac{1}{25} + \frac{1}{10} \right)$$

$$m = 14$$

22. **Ans (1)**

$$\frac{1}{v} - \frac{1}{-f} = \frac{1}{f}$$

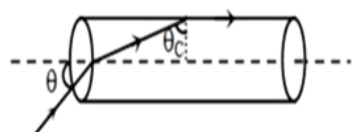
$$\frac{1}{v} = \frac{1}{f} - \frac{2}{f} = -\frac{1}{f}$$

$$\text{or } v = -f$$

$$\text{Again } m = \frac{v}{u} = \frac{-f}{-\frac{f}{2}} = 2$$

Clearly, the image is virtual and doubles the size

23. **Ans (4)**



From the Snell's law,

$$\mu_{\text{air}} \sin \theta = \mu_{\text{cylinder}} \sin(90 - \theta_c)$$

$$1 \times \sin \theta = \frac{2}{\sqrt{3}} \times \cos \theta_c$$

$$\text{but, } \sin \theta_c = \frac{1}{\mu_{\text{cylinder}}} = \frac{\sqrt{3}}{2}$$

$$\text{therefore, } \cos \theta_c = \frac{1}{2}$$

$$\sin \theta = \frac{2}{\sqrt{3}} \cdot \frac{1}{2} = \frac{1}{\sqrt{3}}$$

$$\theta = \sin^{-1} \frac{1}{\sqrt{3}}$$

24. **Ans (3)**

Focal length of plane convex lens $f_2 = F$

Focal length of plane surface mirror f_m

$$P = \frac{2}{f_2} + \frac{1}{f_m}$$

$$P = \frac{2}{F} + \frac{1}{\infty}$$

$$P = \frac{2}{F}$$

25. **Ans (2)**

According to magnifying power,

$$\beta = \frac{\text{angle made by image}}{\text{angle made by object}}$$

As $\beta > 1$ visual angle increases.

26. **Ans (1)**

$$\text{In air } \delta = (1.5 - 1) A = \frac{A}{2}$$

In water

$$\delta = \left(\frac{\mu_g}{\mu_w} - 1 \right) A = \left(\frac{1.5}{1.33} - 1 \right) A = \frac{A}{8}$$

$$\frac{\delta_{\text{in water}}}{\delta_{\text{in air}}} = \frac{1}{4}$$

27. **Ans (4)**

$$f_0 = 1.5 \text{ cm}, f_e = 6.25 \text{ cm}$$

$$u_0 = -D = -25 \text{ cm}$$

By objective lens

$$\frac{1}{f_0} = \frac{1}{v_0} - \frac{1}{u_0}$$

$$\frac{1}{v_0} = \frac{1}{1.5} - \frac{1}{2}$$

$$v_0 = 6 \text{ cm}$$

By eyepiece lens

$$\frac{1}{f_e} = \frac{1}{v_e} - \frac{1}{u_e}$$

$$\frac{1}{6.25} = \frac{1}{-25} - \frac{1}{-u_e}$$

$$\frac{1}{u_e} = \frac{1}{6.25} + \frac{1}{25}$$

$$\frac{1}{u_e} = \frac{4}{25} + \frac{1}{25}$$

$$\frac{1}{u_e} = \frac{1}{5} \Rightarrow u_e = 5$$

Length of tube

$$L = v_0 + u_e = 6 + 5 = 11 \text{ cm}$$

28. **Ans (2)**

Radius of the circular horizon is

$$r = \frac{h}{\sqrt{\mu^2 - 1}}$$

$$r = \frac{12}{\sqrt{\left(\frac{4}{3}\right)^2 - 1}}$$

$$r = \frac{36}{\sqrt{7}} \text{ cm}$$

29. **Ans (3)**

The deviation of light is maximum in case of a convex mirror which provides a greater field of view.

30. **Ans (2)**

$$\sin C = \frac{\mu_R}{\mu_D} = \frac{\lambda_D}{\lambda_R} = \frac{4000}{6000} = \frac{2}{3}$$

31. **Ans (2)**

$$\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2} - \frac{d}{f_1 f_2}$$

$$\therefore \frac{1}{\infty} = \frac{1}{20} + \frac{1}{-5} - \frac{d}{20(-5)}$$

$$\Rightarrow 0 = \frac{1}{20} - \frac{1}{5} + \frac{d}{100}$$

$$\therefore \frac{d}{100} = \frac{1}{5} - \frac{1}{20} = \frac{20-5}{100}$$

$$\therefore d = 15 \text{ cm}$$

32. **Ans (4)**

Magnifying power for normal adjustment (distant object) is given by,

$$M.P. = \frac{\beta}{\alpha} = \frac{f_0}{f_e}$$

$$\Rightarrow \frac{\beta}{2} = \frac{60}{5}$$

$$\Rightarrow \beta = 24^\circ$$

33. **Ans (3)**

Wavelength of red light is more than violet light. Refractive index of a medium is different for light rays of different wave lengths. Larger the wavelengths, the lesser are refractive index.

Now angle of minimum deviation increases with increment in refractive index,

$$\delta = (\mu - 1)A$$

Angle of minimum deviation for red light is less than the violet.

34. **Ans (3)**

$$m = 5 = -\frac{v}{u} \Rightarrow v = -5u$$

$$f = -0.4 \text{ m}$$

From mirror formula

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

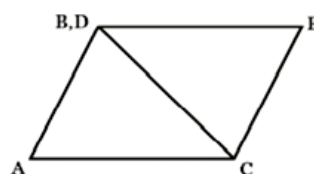
$$\frac{1}{-5u} + \frac{1}{u} = \frac{1}{-0.4}$$

$$\frac{-1+5}{5u} = \frac{1}{-0.4}$$

$$u = \frac{4}{5}(-0.4) = -0.32 \text{ m};$$

So distance from object from mirror = 0.32 m

35. **Ans (1)**



In the position shown net deviation suffered by the ray of light should be minimum. Therefore, the second prism should be rotated by 60° (anticlockwise).

36. **Ans (2)**

For eye piece

$$\frac{1}{V_1} - \frac{1}{u} = \frac{1}{f}$$

$$\frac{-1}{25} - \frac{1}{u} = \frac{1}{5}$$

$$u = -\frac{25}{6}$$

$$m_e = \frac{-25 \times 6}{-25} = 6 \quad (m = m_o m_e = 30)$$

$$m_o = 5$$

37. Ans (1)

An optical fiber is a transparent thin fiber, usually made of glass or plastic for transmitting light. Optical fibers are used in image optics, and work on the principle of total internal reflection of light. Bundles of fibers are used along with the lenses for long, thin image devices called endoscope, which are used to view objects through a small hole. Medical endoscope is used for minimally invasive exploratory or surgical procedure (endoscopy).

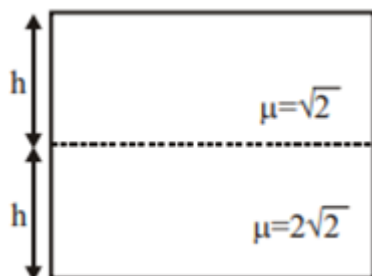
38. Ans (1)

Critical angle $\theta_C = \sin^{-1} \frac{1}{\mu}$

Wavelength increases in the sequence of VIBGYOR. According to Cauchy's formula refractive index decreases as the wavelength increases. Hence the refractive index will increase in the sequence of ROYGBIY.

The critical angle θ_c will thus increase in the same order VIBGYOR. For green light, the incidence angle is just equal to the critical angle. For yellow, orange and red the critical angle will be greater than the incidence angle. So, these colours will emerge from glass-air interface. Hence, the correct option is (yellow, orange, red)

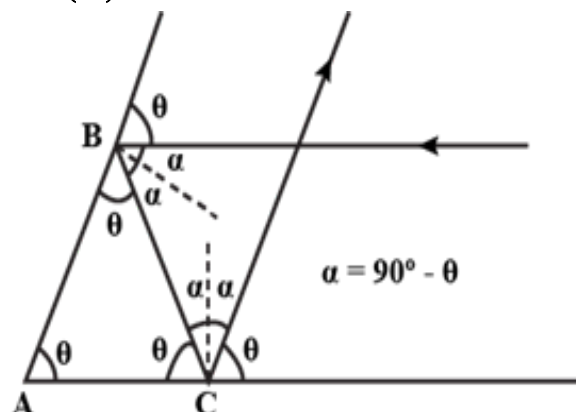
39. Ans (4)



$$d = \frac{h}{\sqrt{2}} + \frac{h}{2\sqrt{2}}$$

$$d = \frac{h}{\sqrt{2}} \times \frac{3}{2} = \frac{3\sqrt{2}h}{4}$$

40. Ans (2)



From the ray diagram-

$$\alpha = 90^0 - \theta$$

$$3\theta = 180^\circ$$

$$\theta = 60^\circ$$

41. Ans (2)

$$\mathbf{P}_1 + \mathbf{P}_2 = 9$$

$$P = P_1 + P_2 - dP_1P_2$$

$$\frac{27}{5} = 9 - \frac{20}{100} \times P_1 P_2$$

The above equation is correct only for

$$P_1 = 3 \text{ and } P_2 = 6$$

42. Ans (4)

The effective focal length of the silvered lens is given by

$$\frac{1}{F} = -\frac{2}{f} + \frac{1}{f_m} = -\frac{2}{15} + \frac{1}{\infty} = -\frac{2}{15}$$

Which gives $F = -\frac{15}{2}$ cm

The silvered lens behaves like a concave mirror. Using the spherical mirror formula

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

we have ($u = -20$ cm and $F = -15/2$ cm)

$$\frac{1}{v} + \frac{1}{-20} = \frac{2}{-15}$$

Which gives $v = -12$ cm. the negative sign indicate that the image is formed to the left of the lens

43. Ans (3)

The concave mirror formula is

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f} \quad \dots(i)$$

Since for a given concave mirror, focal length f is fixed, we partially differentiate. Eq (i) to get

$$-\frac{\delta v}{v^2} - \frac{\delta u}{u^2} = 0 \text{ or } \delta v - \left(\frac{v}{u}\right)^2 \delta u \quad \dots(ii)$$

Multiplying eq (i) by u we get

$$\frac{u}{v} + 1 = \frac{u}{f} = \frac{f}{u-f} \quad \dots(iii)$$

Using eq (iii) in equation (ii) we get

$$\delta v = - \left(\frac{f}{u-f}\right)^2 \delta u$$

Given $\delta u = b$. Therefore

$$\delta v = - \left(\frac{f}{u-f}\right)^2 b$$

The negative sign shown the image is longitudinally inverted. The magnitude of the size of the image is

$$\delta v = \left(\frac{f}{u-f}\right)^2 \delta u$$

44. Ans (1)

$$(\mu_{v1} - \mu_{r1})\alpha_1 = (\mu_{v2} - \mu_{r2})\alpha_2$$

$$\alpha_2 = \frac{1.523 - 1.513}{1.665 - 1.645} \times 6^\circ = 3^\circ$$

45. Ans (1)

$$\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$$

$$\frac{1.5}{100} + \frac{1}{100} = \frac{1.5 - 1}{R}$$

$$R = 20 \text{ cm}$$